ECOVISION ACADEMY

STATA

COMPLETE STATA
(ECONOMETRICS & DATA ANALYSIS) PROGRAMME

- Course Fee: ₹399 Only
- Total Duration: 30 Hours
- Level: Beginner to Advanced
- Outcome: Research-Ready STATA + Econometric Skills + Academic & Policy Analysis Confidence

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TRAINING CURRICULUM

SECTION 1: Introduction to Data Analysis & STATA
Duration: 1 Hour

- Role of Data Analysis in Economics & Social Sciences
- Introduction to STATA Software
- Where STATA is Used (Research, Policy, IMF, World Bank, RBI)
- STATA vs SPSS vs R vs Python
- Understanding STATA Workflow

SECTION 2: STATA Installation & Interface
Duration: 1 Hour

- Installing STATA
- Understanding Command Window, Results, Variables
- Do-file Editor & Log Files
- Basic Navigation & Shortcuts

SECTION 3: Data Types & Data Structure in STATA
Duration: 1.5 Hours

- Cross-Sectional, Time Series & Panel Data
- Numeric & String Variables
- Variable Labels & Value Labels
- Understanding Missing Values
- Inspecting Data Structure

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4 SECTION 4: Importing Data into STATA
Duration: 1.5 Hours

- Importing Excel, CSV & Text Files
- Importing SPSS & SAS Data
- Using import excel & use commands
- Saving & Managing STATA Datasets

5 SECTION 5: Data Cleaning & Management
Duration: 2 Hours

- Renaming Variables
- Dropping & Keeping Observations
- Handling Missing Values
- Generating New Variables
- Data Transformation Techniques
- Cleaning Real Research Data

6 SECTION 6: Desciptive Statistics & Summary Analysis
Duration: 1.5 Hours

- Measures of Central Tendency
- Measures of Dispersion
- summarize, tabulate, describe
- Group-wise Descriptive Statistics
- Exporting Tables for Reports

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7 SECTION 7: Data Visualization in STATA
Duration: 1.5 Hours

- Line Graphs & Scatter Plots
- Bar Charts & Histograms
- Box Plots
- Customizing Graphs
- Exporting Graphs for Papers

8 SECTION 8: Correlation & Association Analysis
Duration: 1 Hour

- Covariance & Correlation
- Pearson & Spearman Correlation
- Correlation Matrix
- Interpretation for Research

9 SECTION 9: Probability Distributions & Sampling
Duration: 1 Hour

- Normal Distribution
- Binomial & Poisson Distribution
- Random Sampling
- Simulation Basics in STATA

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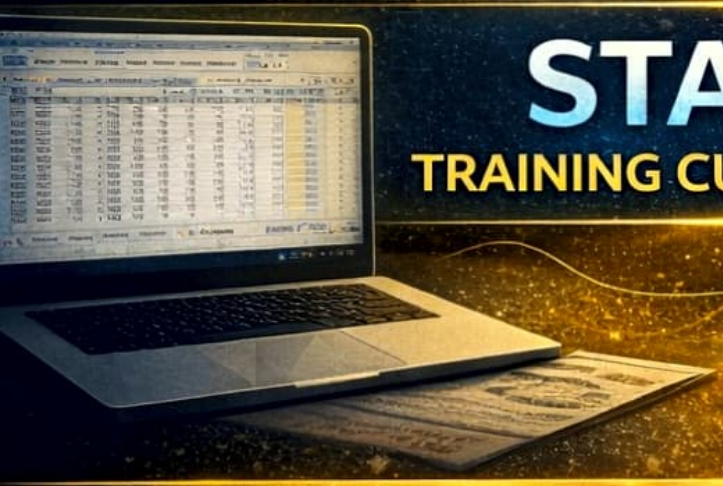

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10 SECTION 10: Hypothesis Testing
Duration: 2 Hours

- Null & Alternative Hypothesis
- Z-test & t-test
- Paired vs Independent Samples
- ANOVA
- Interpreting p-values & Confidence Intervals

11 SECTION 11: Simple & Multiple Regression Analysis
Duration: 2 Hours

- Concept of Regression
- Ordinary Least Squares (OLS)
- Interpretation of Coefficients
- Goodness of Fit (R^2)
- Real-World Examples

12 SECTION 12: Regression Diagnostics
Duration: 1.5 Hours

- Multicollinearity
- Heteroskedasticity
- Autocorrelation
- Model Specification Errors
- Remedies & Robust Estimation

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13 SECTION 13: Dummy Variables & Interaction Terms
Duration: 1 Hour

- Creating Dummy Variables
- Categorical Variables in Regression
- Interaction Effects
- Policy Interpretation
- Interpreting p-values & Confidence Intervals

14 SECTION 14: Time Series Analysis
Duration: 2 Hours

- Time Series Data Structure
- Trend & Seasonality
- Stationarity & Unit Root Test
- AR, MA, ARIMA, ARIMA Models
- Forecasting Basics

15 SECTION 15: Panel Data Analysis
Duration: 2 Hours

- Concept of Panel Data
- Fixed Effects Model
- Random Effects Model
- Hausman Test
- Applications in Development Economics

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16 SECTION 16: Limited Dependent Variable Models
Duration: 1.5 Hours

- Logit Model
- Probit Model
- Interpretation of Marginal Effects
- Binary Outcome Analysis

17 SECTION 17: Advanced Econometric Topics
Duration: 1.5 Hours

- Instrumental Variable (IV) Regression
- Two Stage Least Squares (2SLS)
- Difference-in-Differences (DID)
- Basic Causal Inference

18 SECTION 18: Working with Real-World Datasets
Duration: 1.5 Hours

- NSSO / PLFS Data Handling
- World Bank & IMF Data
- Policy & Research Datasets
- Data Preparation for Thesis

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19 SECTION 19: Automating Work Using Do-Files
Duration: 1 Hour

- Writing Do-Files
- Reproducible Research
- Loops & Macros (Basics)
- Efficient Workflow

21 SECTION 20: STATA Mini Research Projects
Duration: 2 Hours

- Growth & Development Analysis
- Education & Labour Market Study
- Poverty & Inequality Analysis
- Policy Evaluation Project

21 SECTION 21: STATA Mini Research Projects
Duration: 2 Hours

- Growth & Development Analysis
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22 SECTION 22: STATA FOR EXAMS, RESEARCH & CAREERS

Duration: 1 Hour

- STATA for NET / SET / PhD
- Common Research Mistakes
- Interview & Viva Questions
- Career Path in Research & Analytics

23 SECTION 23: FINAL ASSESSMENT & CERTIFICATION

Duration: 1 Hour

- Final Practical Test
- Project Evaluation
- Certification Criteria
- Research Career Roadmap

23 SECTION 23: FINAL ASSESSMENT & CERTIFICATION

Duration: 1 Hour